

PooleQ/P Capture: Channel-Resolved Hysteretic Defect Logic and Decomposed Geometry Spectrometry for p-bit Fields and Quantum Syndrome Data

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Abstract

This amended v5.6 note keeps PooleQ/P Capture within a strict measurement-side boundary while upgrading its interpretation from a single capture-score framework into a local-geometry spectrometer for measured binary fields. The base Poole update on a three-dimensional Moore lattice is

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\},$$

where N_i is the 26-neighbor active count. The central move is to preserve birth, survival, over-support, and strain channels instead of collapsing the update immediately into one live/dead output. Earlier amendments established the p-bit probability layer, exact-cardinality defect footprints, biased Hamming-window gates, channel-resolved adder readouts, the rectangular defect-transducer boundary, same-density shuffled controls, and the v5.5 decomposed robust scores. The v5.6 amendment names the resulting structure: PooleQ/P acts as a geometry spectrometer with signature vector

$$\mathcal{S}_{\mathcal{P}}(X) = (R_B(X), R_H(X), R_{\psi}(X)),$$

where R_B measures birth-window/boundary geometry, R_H measures high-support geometry, and R_{ψ} measures support-strain geometry. In the v5.5 verification run used by this amendment, every tested field had the same active density, 0.059995; the maximum combined robust score among IID and shuffled controls was 1.0959, while the minimum structured score was 4.5899. Thus the paper-safe v5.6 claim is: PooleQ/P maps measured binary fields into an interpretable local-geometry signature space that separates structured fields from exact same-density shuffled controls and indicates which local geometry carries the separation. It remains a classical transform for p-bit samples, readout strings, and syndrome histories; it does not copy, clone, or reconstruct an unknown qubit state.

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1 Purpose and boundary

The Poole Manifold rule was originally considered as a three-dimensional binary cellular automaton with a 26-neighbor Moore neighborhood. Previous work showed that it acts as a hysteretic surface selector: one-cell-thick sheets are preserved, bulk is rejected, and surface defects generate normal-layer birth halos. This note asks a narrower mathematical and computational question:

Can the channel structure of the Poole rule be used as a local computational capture transform for stochastic p-bit fields and measured qubit syndrome fields?

The answer developed here is yes, with a strict boundary. PooleQ/P Capture does *not* capture the private state of an unknown qubit. Unknown quantum states cannot be cloned or perfectly copied by a physical operation [1, 2]. The target data are already-measured classical fields:

- p-bit samples $X_i(t) \in \{0, 1\}$ or p-bit probabilities $p_i(t) \in [0, 1]$;
- measured qubit readout strings $m_i(t) \in \{0, 1\}$;
- stabilizer syndrome fields or syndrome-event histories $S(x, y, t) \in \{0, 1\}$.

In this setting, PooleQ/P is a classical transform applied after measurement. Its proposed role is comparable to a feature map, local decoder preprocessor, or defect-sensitive stochastic layer. It should not be described as a method for reconstructing unmeasured quantum amplitudes.

2 The Poole Manifold rule

Let $V = (\mathbb{Z}/L\mathbb{Z})^3$ be a periodic three-dimensional lattice. A binary state is a map

$$x : V \rightarrow \{0, 1\}.$$

For $i \in V$, let $\mathcal{N}_{26}(i)$ denote the 26-cell Moore neighborhood:

$$\mathcal{N}_{26}(i) = \{i + \delta : \delta \in \{-1, 0, 1\}^3 \setminus \{0\}\}.$$

The local support count is

$$N_i(x) = \sum_{j \in \mathcal{N}_{26}(i)} x(j).$$

The Poole update is

$$\mathcal{P}(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}. \quad (1)$$

Equivalently, inactive cells are born for $N_i \in \{5, 6, 7\}$, while active cells survive for $N_i \in \{5, 6, 7, 8, 9\}$.

Proposition 1 (Compact hysteretic form). *The rule B5-7/S5-9 is exactly equivalent to Eq. (1).*

Proof. If $x(i) = 0$, the upper bound in Eq. (1) is 7, so $\mathcal{P}(x)_i = 1$ exactly when $5 \leq N_i \leq 7$. If $x(i) = 1$, the upper bound is 9, so $\mathcal{P}(x)_i = 1$ exactly when $5 \leq N_i \leq 9$. $\square$

The term $2x(i)$ is the hysteretic memory term: the prior active state gives the cell a two-neighbor upper-support allowance. This is what allows a smooth one-cell-thick sheet cell with $N_i = 8$ to survive.

3 Channel-resolved Poole capture

The ordinary update collapses all allowed birth and survival events into the same binary output. For capture, the local channels are kept separated.

Definition 1 (Birth and survival channels). *For $k = 0, 1, \dots, 26$, define*

$$\mathbf{B}_k(i; x) = (1 - x(i))\mathbf{1}\{N_i(x) = k\}, \quad (2)$$

$$\mathbf{S}_k(i; x) = x(i)\mathbf{1}\{N_i(x) = k\}. \quad (3)$$

The active Poole channels are $\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7$ and $\mathbf{S}_5, \mathbf{S}_6, \mathbf{S}_7, \mathbf{S}_8, \mathbf{S}_9$.

Then

$$\mathcal{P}(x)_i = \sum_{k=5}^7 \mathbf{B}_k(i; x) + \sum_{k=5}^9 \mathbf{S}_k(i; x). \quad (4)$$

Because exactly one value of N_i holds at any site, the terms in Eq. (4) are disjoint.

Definition 2 (PooleQ/P capture vector). *The local PooleQ/P feature vector is*

$$\Phi_i(x) = (\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7, \mathbf{S}_5, \mathbf{S}_6, \mathbf{S}_7, \mathbf{S}_8, \mathbf{S}_9, O_{10+}, \psi)_i, \quad (5)$$

where

$$O_{10+}(i; x) = x(i) \mathbf{1}\{N_i(x) \geq 10\}$$

and

$$\psi_i(x) = \max\{0, N_i(x) - (7 + 2x(i))\} - \max\{0, 5 - N_i(x)\}. \quad (6)$$

Here $\psi < 0$ is under-support, $\psi = 0$ is valid Poole support, and $\psi > 0$ is over-support.

Remark 1 (The observability issue). *The channel-resolved transform $\Phi_i(x)$ is richer than the raw automaton state $\mathcal{P}(x)_i$. If only the next binary state is observed, the three birth channels $\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7$ become indistinguishable. Thus several results below are exact for typed readout, PooleScan-style feature readout, or spatially demultiplexed gate layouts. They are not automatically exact for a one-state, uninstrumented CA implementation.*

4 Probabilistic p-bit form

A p-bit is a stochastic binary variable that fluctuates between 0 and 1 with a controlled probability. p-bit systems have been proposed as probabilistic computing substrates and as a bridge between ordinary bits and qubit-inspired hardware [3, 4].

Let

$$X_j \sim \text{Bernoulli}(p_j)$$

independently over $j \in V$. Since the center cell X_i is not included in N_i , the center variable is independent of the local neighbor count under this independent-site model.

4.1 Poisson-binomial channel probabilities

For a fixed site i , define the local generating function

$$G_i(z) = \prod_{j \in \mathcal{N}_{26}(i)} (1 - p_j + p_j z). \quad (7)$$

Then

$$\mathbb{P}[N_i(X) = k] = [z^k] G_i(z). \quad (8)$$

This is a local Poisson-binomial distribution.

Proposition 2 (Expected Poole channels). *Under the independent Bernoulli model,*

$$\mathbb{E}[\mathbf{B}_k(i; X)] = (1 - p_i) [z^k] G_i(z), \quad (9)$$

$$\mathbb{E}[\mathbf{S}_k(i; X)] = p_i [z^k] G_i(z). \quad (10)$$

Therefore the expected Poole activation is

$$q_i := \mathbb{E}[\mathcal{P}(X)_i] = (1 - p_i) \sum_{k=5}^7 [z^k] G_i(z) + p_i \sum_{k=5}^9 [z^k] G_i(z). \quad (11)$$

Proof. Equation (8) follows from the generating function for a sum of independent non-identical Bernoulli variables. Since X_i is not a factor in N_i , X_i is independent of N_i , giving Eqs. (9) and (10). Summing the active channels gives Eq. (11). $\square$

Define

$$C_i^{a:b} = \sum_{k=a}^b [z^k] G_i(z). \quad (12)$$

Then

$$q_i = (1 - p_i) C_i^{5:7} + p_i C_i^{5:9}. \quad (13)$$

Because $C_i^{5:9} = C_i^{5:7} + C_i^{8:9}$, this can be written as

$$q_i = C_i^{5:7} + p_i C_i^{8:9}. \quad (14)$$

This is the probabilistic form of hysteretic memory: the center probability p_i only weights the upper-memory band 8, 9.

4.2 Differentiability

For $\ell \in \mathcal{N}_{26}(i)$,

$$\frac{\partial}{\partial p_\ell} [z^k] G_i(z) = [z^k] \left((z - 1) \prod_{j \in \mathcal{N}_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right). \quad (15)$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}, \quad (16)$$

again showing that the center probability only gates the memory band.

4.3 Poole-biased p-bit dynamics

A p-bit network can use channel samples or expected channels as features. One possible stochastic update is

$$\mathbb{P}[X_i(t+1) = 1 \mid X_t] = \sigma \left(\beta \left[\theta_i + w^\top \Phi_i(X_t) \right] \right), \quad (17)$$

where $\sigma(u) = (1 + e^{-u})^{-1}$. A differentiable probability-layer version replaces $\Phi_i(X_t)$ by $\mathbb{E}[\Phi_i(X_t)]$ under a local product approximation.

5 Correlation residue and null calibration

PooleQ/P can measure local dependence beyond independent-site statistics. Suppose M samples $X^{(1)}, \dots, X^{(M)}$ are observed from a p-bit device or repeated measurement experiment. Define empirical channel rates

$$\hat{B}_k(i) = \frac{1}{M} \sum_{m=1}^M B_k(i; X^{(m)}), \quad \hat{S}_k(i) = \frac{1}{M} \sum_{m=1}^M S_k(i; X^{(m)}). \quad (18)$$

Let

$$\hat{p}_i = \frac{1}{M} \sum_{m=1}^M X_i^{(m)}. \quad (19)$$

Using empirical marginals $\hat{p}_j$, form the independent prediction q_i^{ind} from Eq. (11). The scalar Poole activation residue is

$$\Delta_i = \hat{q}_i - q_i^{\text{ind}}, \quad \hat{q}_i = \frac{1}{M} \sum_{m=1}^M \mathcal{P}(X^{(m)})_i. \quad (20)$$

A channel-wise residue vector is

$$R_i = \left(\hat{\mathbf{B}}_5 - \mathbb{E}_{\text{ind}} \mathbf{B}_5, \hat{\mathbf{B}}_6 - \mathbb{E}_{\text{ind}} \mathbf{B}_6, \hat{\mathbf{B}}_7 - \mathbb{E}_{\text{ind}} \mathbf{B}_7, \hat{\mathbf{S}}_8 - \mathbb{E}_{\text{ind}} \mathbf{S}_8, \hat{\mathbf{S}}_9 - \mathbb{E}_{\text{ind}} \mathbf{S}_9 \right)_i. \quad (21)$$

Definition 3 (Poole coherence). *A simple global correlation statistic is*

$$\mathcal{C}_{\mathcal{P}} = \sum_{i \in V} \|R_i\|_2^2. \quad (22)$$

A large value of $\mathcal{C}_{\mathcal{P}}$ indicates that local Poole channel rates are not explained by the product of observed local marginals. It does not by itself prove quantum entanglement or a particular physical noise model. It is a local structured-correlation detector.

6 Defect cardinality logic

Around a stable Poole sheet, the channel-resolved birth rule becomes exact-cardinality logic. Let S be a one-cell-thick coordinate sheet in the $z = 0$ plane. A perfect sheet is an exact Poole fixed point: active sheet cells have $N = 8$, and inactive cells immediately above or below the sheet have $N = 9$, which is above the birth window.

Let D be a set of missing cells in the sheet. For a normal-layer candidate cell directly above or below a sheet location $u \in (\mathbb{Z}/L\mathbb{Z})^2$, define the defect-footprint count

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}. \quad (23)$$

This counts missing cells from the 3×3 sheet footprint below the candidate. For the first update, assuming no pre-existing active cells in the adjacent normal layer, the candidate's active neighbor count is

$$N = 9 - r_D(u). \quad (24)$$

Therefore

$$\mathbf{B}_7 \iff r_D = 2, \quad \mathbf{B}_6 \iff r_D = 3, \quad \mathbf{B}_5 \iff r_D = 4. \quad (25)$$

Proposition 3 (Channel-cardinality equivalence). *For a normal-layer candidate cell adjacent to a defective sheet, under the first-update sheet approximation,*

$$\mathbf{B}_k = 1 \iff r_D = 9 - k, \quad k \in \{5, 6, 7\}.$$

6.1 Biased exact-cardinality detectors

Let $A_1, \dots, A_n \in \{0, 1\}$ be input defects placed at selected footprint positions, and let $b \in \{0, 1, \dots, 9 - n\}$ be the number of fixed bias defects in the same 3×3 footprint. Define

$$s = \sum_{j=1}^n A_j, \quad r = s + b. \quad (26)$$

By Eq. (25), the birth channels detect exact cardinalities:

$$\mathbf{B}_7 = \mathbf{1}\{s = 2 - b\}, \quad \mathbf{B}_6 = \mathbf{1}\{s = 3 - b\}, \quad \mathbf{B}_5 = \mathbf{1}\{s = 4 - b\}. \quad (27)$$

Thus a biased footprint is a small exact-count circuit.

7 Raw biased-footprint Boolean gates

The exact-cardinality channels above require channel-resolved readout. There is also a lower-level fact: the collapsed raw birth event itself is already a useful Boolean gate.

For an inactive normal-layer candidate adjacent to a defective sheet, the raw birth output is

$$Y = B_5 \vee B_6 \vee B_7. \quad (28)$$

By Eq. (25), this is equivalent to

$$Y = \mathbf{1}\{2 \leq r \leq 4\}. \quad (29)$$

If $A_1, \dots, A_n \in \{0, 1\}$ denote input defects and c denotes fixed bias defects in the same footprint, then

$$Y_{n,c}(A_1, \dots, A_n) = \mathbf{1} \left\{ 2 \leq c + \sum_{j=1}^n A_j \leq 4 \right\}. \quad (30)$$

Theorem 1 (Biased defect-footprint logic). *Under the first-update sheet-footprint assumptions, for two inputs A, B ,*

$$Y_{2,0}(A, B) = A \wedge B, \quad (31)$$

$$Y_{2,1}(A, B) = A \vee B, \quad (32)$$

$$Y_{2,3}(A, B) = \neg(A \wedge B), \quad (33)$$

$$Y_{2,4}(A, B) = \neg(A \vee B). \quad (34)$$

Thus the raw Poole defect-footprint birth operator contains functionally complete Boolean primitives.

Proof. If r cells are missing from the 3×3 sheet footprint, then the inactive normal-layer candidate sees $N = 9 - r$ active neighbors. The Poole birth condition $5 \leq N \leq 7$ is equivalent to $2 \leq r \leq 4$. With $r = c + \sum_j A_j$, Eq. (30) follows. For $n = 2$, substitute $c = 0, 1, 3, 4$. $\square$

The same construction gives one-input identity and inversion gates:

$$Y_{1,1}(A) = A, \quad Y_{1,4}(A) = \neg A. \quad (35)$$

Channel-resolved readout is not needed for NAND or NOR; it becomes necessary for arithmetic compression, where one must distinguish whether the birth arose from B_5 , B_6 , or B_7 .

8 Half-adder and full-adder

Let $A, B \in \{0, 1\}$ be two input defects, and set one fixed bias defect $b = 1$. Then

$$r = A + B + 1.$$

From Eq. (27),

$$B_7 = \mathbf{1}\{A + B = 1\} = A \oplus B, \quad (36)$$

$$B_6 = \mathbf{1}\{A + B = 2\} = A \wedge B. \quad (37)$$

Thus SUM = B_7 and CARRY = B_6 .

For a full-adder, let $A, B, C_{\text{in}} \in \{0, 1\}$ and again set $b = 1$. If

$$s = A + B + C_{\text{in}},$$

then

$$B_7 = \mathbf{1}\{s = 1\}, \quad B_6 = \mathbf{1}\{s = 2\}, \quad B_5 = \mathbf{1}\{s = 3\}. \quad (38)$$

The full-adder sum bit is parity, true for $s = 1$ or $s = 3$, and the carry bit is true for $s \geq 2$. Hence

$$\boxed{\text{SUM} = B_7 \vee B_5, \quad \text{CARRY} = B_6 \vee B_5.} \quad (39)$$

Remark 2 (What is exact, and what remains engineering). *The raw biased-footprint theorem and the channel-resolved full-adder theorem are exact local first-update statements, but they do not yet make an autonomous one-state Poole computer. Routing, fanout, timing, reset, gate isolation, and channel demultiplexing remain engineering and verification problems. If only the collapsed output $\mathcal{P}(x)_i \in \{0, 1\}$ is observed, the full-adder footprint collapses to $A \vee B \vee C_{\text{in}}$, not the full-adder output.*

9 Application to qubit syndrome fields

Surface-code quantum computing represents quantum information using many physical qubits and repeated stabilizer measurements; classical decoders infer corrections from syndrome data [5, 6, 7]. PooleQ/P does not replace these decoders. Instead, it can be defined as a local feature layer on binary syndrome spacetime.

Let

$$S(x, y, t) \in \{0, 1\}$$

be a syndrome-event history. Treating (x, y, t) as a three-dimensional lattice, define

$$\Phi(S)(x, y, t) = (B_5, B_6, B_7, S_5, \dots, S_9, O_{10+}, \psi)(x, y, t). \quad (40)$$

This maps a syndrome history into local features describing event-chain thickness, over-support, local crowding, and hysteretic survival windows. A decoder can then be augmented as

$$\hat{E} = D_\theta(S, \Phi(S)), \quad (41)$$

where D_θ may be a matching, union-find, neural, or hybrid decoder. The claim is not that PooleQ/P is a complete decoder by itself. The claim is that it supplies interpretable local geometry features for measured syndrome data.

10 v5.1 scope note: static Q/P capture versus dynamic Defect Calculus

The v5.1 amendment kept PooleQ/P separate from the Defect Calculus paper. Static first-step Q/P capture is not the same as multi-step response capture. Multi-step planar relaxation and 3D void response belong to the Defect Calculus empirical atlas unless a future dynamic Q/P extension explicitly defines the model, measurement channel, and time window.

This boundary is kept in v5.5. The new benchmark below is a measurement-side structured-correlation benchmark. It does not replace the static footprint theorems or the separate dynamic Defect Calculus program.

11 v5 amendment: rectangle transducer boundary

For an $a \times b$ rectangular planar puncture with $a, b \geq 2$, the Defect Calculus v9 paper gives the raw response

$$B_0(a, b) = 4a + 4b + 12, \quad (42)$$

and the default prime-sharpened response

$$B_\phi(a, b) = 4a + 4b - 4. \quad (43)$$

In perimeter form, with $P = 2a + 2b$,

$$B_0 = 2P + 12, \quad B_\phi = 2P - 4. \quad (44)$$

Thus the defect response is a perimeter-like capture channel.

Theorem 2 (Raw Q/P rectangle reconstruction). *For a raw rectangular puncture, area $A = ab$ and raw response B_0 recover the unordered side pair $\{a, b\}$ by*

$$s = \frac{B_0 - 12}{4} = a + b, \quad (45)$$

$$\Delta = s^2 - 4A, \quad (46)$$

$$\{a, b\} = \left\{ \frac{s - \sqrt{\Delta}}{2}, \frac{s + \sqrt{\Delta}}{2} \right\}. \quad (47)$$

Theorem 3 (Prime Q/P rectangle reconstruction). *For a default prime rectangular puncture, area A and prime response B_ϕ recover the unordered side pair by*

$$s = \frac{B_\phi + 4}{4} = a + b, \quad (48)$$

with the same discriminant formula.

Remark 3 (Corrected reconstruction boundary). *Birth count alone is not a complete rectangle reconstruction invariant. It determines $a + b$ but not ab . The corrected claim is therefore: area plus response reconstructs rectangle dimensions up to exchange. This is a prior-dependent rectangle theorem, not a claim that global counts reconstruct arbitrary masks.*

12 v5.4/v5.5 amendment: null-calibrated and decomposed capture

This amendment turns the earlier correlation-residue idea into a direct empirical benchmark for measured binary fields. The target is not private quantum-state recovery. The target is structured correlation in already-measured binary data.

Let

$$X^{(1)}, \dots, X^{(M)} \in \{0, 1\}^V$$

be repeated measured binary fields. For example, these may be p-bit samples, measured qubit readout strings, or syndrome-event histories. Define the empirical channel-rate vector

$$\hat{\mu}_\Phi = \frac{1}{M} \sum_{m=1}^M \text{rate}_V[\Phi(X^{(m)})]. \quad (49)$$

12.1 Same-density IID and shuffled controls

Let

$$\hat{p} = \frac{1}{M|V|} \sum_{m=1}^M \sum_{i \in V} X_i^{(m)}$$

be the observed active density. The homogeneous same-density IID null is

$$X_i^{\text{null}} \sim \text{Bernoulli}(\hat{p}), \quad i \in V,$$

independent over sites. In the synthetic benchmark, all classes are also forced to have exactly the same active count. This prevents the score from separating classes merely because one class has more active cells.

For each structured field X , define a shuffled control X^{shuf} by randomly permuting the voxel values. Then

$$\sum_i X_i^{\text{shuf}} = \sum_i X_i,$$

but local geometry is destroyed. If a score is high for X and collapses for X^{shuf} , the score is detecting local arrangement rather than active density.

12.2 Raw and robust capture scores

For a selected channel set $\mathcal{K}$, let μ_0 and σ_0 be the empirical same-density IID mean and standard deviation of the channel-rate vector. The raw null-calibrated capture score is

$$Z_{\mathcal{K}}(X) = \left\| \frac{\hat{\mu}_{\Phi_{\mathcal{K}}}(X) - \mu_0}{\sigma_0 + \varepsilon} \right\|_2. \quad (50)$$

Because rare channels such as S_9 and O_{10+} may have near-zero IID variance in sparse fields, v5.5 continues to report a softened robust score

$$R_{\mathcal{K}}(X) = \left(\sum_{k \in \mathcal{K}} \text{asinh}^2 \left[\frac{1}{\lambda} \frac{\hat{\mu}_{\Phi_k}(X) - \mu_{0,k}}{\sigma_{0,k} + \varepsilon} \right] \right)^{1/2}. \quad (51)$$

In the v5.5 run, $\lambda = 5$. The raw score remains useful as a detector. The robust score is safer for comparison and plotting when a structured class activates channels that are nearly absent under the null.

12.3 Decomposed PooleQ/P score groups

The v5.5 amendment decomposes capture into three interpretable components:

$$\mathcal{K}_B = \{B_5, B_6, B_7\}, \quad (52)$$

$$\mathcal{K}_H = \{S_8, S_9, O_{10+}\}, \quad (53)$$

$$\mathcal{K}_{\psi} = \{\psi_{\text{abs}}\}. \quad (54)$$

Here $\mathcal{K}_B$ measures birth-window or boundary geometry, $\mathcal{K}_H$ measures high-support geometry, and $\mathcal{K}_{\psi}$ measures support-strain geometry. The non-redundant combined set is

$$\mathcal{K}_C = \mathcal{K}_B \cup \mathcal{K}_H \cup \mathcal{K}_{\psi}. \quad (55)$$

The collapsed update rate P_{active} is still reported as a descriptive statistic, but it is excluded from $\mathcal{K}_C$ because it overlaps with the individual birth and survival channels.

For non-null classes, geometry shares are defined from the robust components by

$$s_g = \frac{R_g^2}{R_B^2 + R_H^2 + R_\psi^2 + \varepsilon}, \quad g \in \{B, H, \psi\}. \quad (56)$$

Classes with $R_C < 2$ are labeled null-like before interpreting shares, because a percentage decomposition of a near-zero signal is not meaningful.

13 v5.3 shuffled-control benchmark

The v5.3 Colab-safe benchmark used $L = 28$, $p = 0.06$, $M = 50$ samples per class, seed 13, and exact active count 1317 per field. It established the same-density shuffled-control pattern: structured line, chain, burst, and sheet ensembles had high capture scores, while their exact same-density shuffled controls returned to the IID baseline.

Class	Active density	Poole active	Raw $Z_{\mathcal{P}}$	Robust $R_{\mathcal{P}}$
IID null	0.059995	0.017413	1.71	0.34
Lines	0.059995	0.066732	87.94	5.88
Lines shuffled	0.059995	0.017239	1.67	0.33
Random-walk chains	0.059995	0.081722	215.05	8.36
Walks shuffled	0.059995	0.017360	1.90	0.38
Branching chains	0.059995	0.076872	141.35	7.01
Branches shuffled	0.059995	0.017408	1.67	0.33
Compact bursts	0.059995	0.060024	555.14	9.50
Bursts shuffled	0.059995	0.017602	1.89	0.37
Sheets	0.059995	0.075381	657.61	10.04
Sheets shuffled	0.059995	0.017513	1.82	0.36

Table 1: v5.3 same-density shuffled-control benchmark. Every class has the same active density to six decimal places. Structured classes have high capture scores; shuffled controls return to the IID baseline.

This result justified moving from a single score to the decomposed v5.5 score groups above.

14 v5.5 decomposed benchmark results

The v5.5 run used $L = 28$, $p = 0.06$, $M = 50$, seed 23, and exact active count 1317 per field. The verification report passed all checks: same-density generation, active-count audit, shuffle audit, clean robust separation, and structured-beats-shuffled controls. The maximum combined robust score among IID and shuffled controls was 1.0959, while the minimum combined robust score among structured samples was 4.5899.

Class	Active	R_B	R_H	R_ψ	R_C	Signature
IID null	0.059995	0.21	0.02	0.16	0.28	null-like
Straight lines	0.059995	3.49	2.19	3.28	5.28	mixed geometry
Lines shuffled	0.059995	0.24	0.03	0.17	0.31	null-like
Random-walk chains	0.059995	3.62	5.63	4.15	7.88	high-support-led
Walks shuffled	0.059995	0.23	0.02	0.19	0.31	null-like
Branching chains	0.059995	3.87	3.66	3.88	6.61	mixed geometry
Branches shuffled	0.059995	0.21	0.02	0.14	0.27	null-like
Compact bursts	0.059995	3.12	7.06	5.00	9.19	high-support-led
Bursts shuffled	0.059995	0.23	0.02	0.16	0.29	null-like
Sheets	0.059995	3.50	7.40	5.31	9.76	high-support-led
Sheets shuffled	0.059995	0.22	0.04	0.16	0.29	null-like

Table 2: v5.5 decomposed robust capture scores. R_B uses $\{B_5, B_6, B_7\}$, R_H uses $\{S_8, S_9, O_{10+}\}$, R_ψ uses ψ_{abs} , and R_C uses their non-redundant union. The collapsed update rate P_{active} is reported in the result files but is not included in R_C .

Structured class	R_C	Shuffled R_C	Drop	Signature
Straight lines	5.28	0.31	4.97	mixed geometry
Random-walk chains	7.88	0.31	7.57	high-support-led
Branching chains	6.61	0.27	6.34	mixed geometry
Compact bursts	9.19	0.29	8.90	high-support-led
Sheets	9.76	0.29	9.47	high-support-led

Table 3: Paired same-density shuffled-control results from v5.5. Shuffling preserves active count but destroys local geometry. Every structured class remains high while its shuffled control collapses to the null-like range.

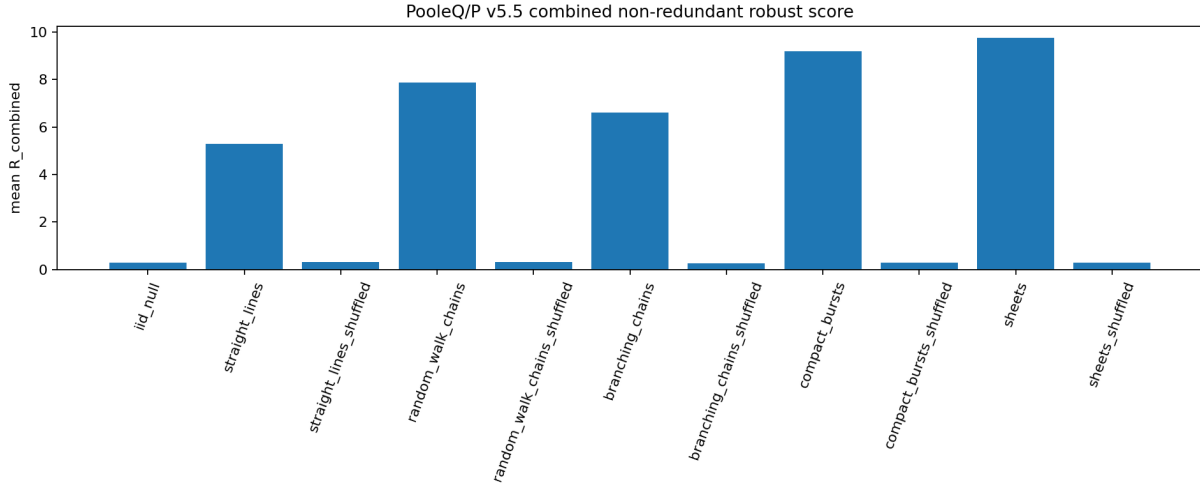


Figure 1: Combined non-redundant robust score R_C for v5.5. Structured fields separate from IID and same-density shuffled controls.

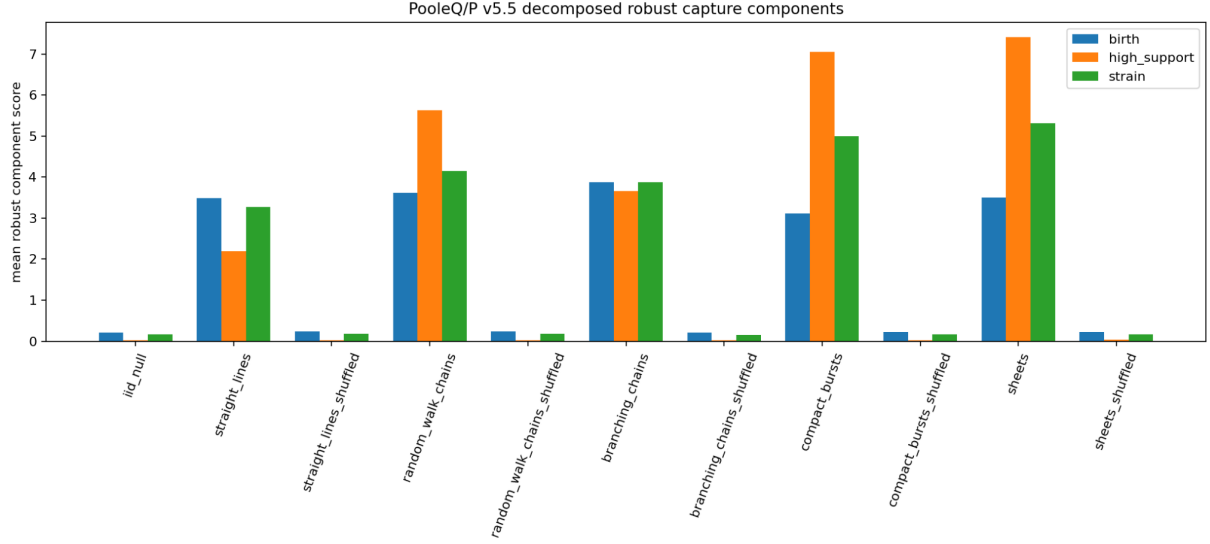


Figure 2: Decomposed robust components. Straight lines and branching chains are mixed; random-walk chains, compact bursts, and sheets are high-support-led.

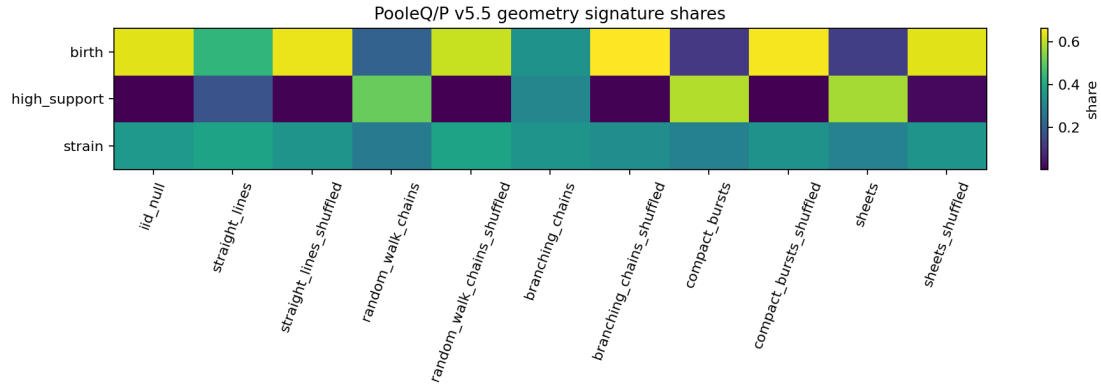


Figure 3: Geometry signature shares. Shares should be interpreted only for non-null classes; IID and shuffled controls are first labeled null-like because their total robust score is below threshold.

14.1 Single-channel interpretation

The strongest single-channel effects agree with the class geometry. Values in Table 4 are mean null-calibrated channel effects and should be treated as diagnostics rather than physical sigma claims.

Class	Top mean channel effects, largest first
Straight lines	ψ_{abs} : 66.8; B_6 : 26.1; B_7 : 23.7; S_8 : 15.6; B_5 : 8.2
Random-walk chains	ψ_{abs} : 159.2; S_8 : 82.6; O_{10+} : 64.1; S_9 : 52.8; B_7 : 33.1
Branching chains	ψ_{abs} : 122.1; B_7 : 40.0; S_8 : 31.9; B_6 : 25.0; O_{10+} : 17.4
Compact bursts	ψ_{abs} : 370.8; O_{10+} : 358.8; S_9 : 91.0; S_8 : 83.4; B_7 : 26.4
Sheets	ψ_{abs} : 508.9; S_8 : 328.9; O_{10+} : 147.7; S_9 : 119.7; B_6 : 28.6

Table 4: Top single-channel effects from the v5.5 benchmark.

The resulting geometry signatures are intuitive. Straight lines are mixed, with birth-window and strain components both active. Branching chains are also mixed, consistent with split local support. Random-walk chains, compact bursts, and sheets are high-support-led, consistent with local bunching, dense support, and over-support channels.

15 v5.6 interpretation: Poole geometry spectrometer

The v5.5 decomposition suggests a cleaner identity for PooleQ/P. The transform should not be described only as a detector that answers whether a field is structured. It is better described as a local-geometry spectrometer: it maps a measured binary field into a low-dimensional signature of the local geometry that produced the channel spectrum.

Definition 4 (Poole geometry signature vector). *For a measured binary field X , define*

$$\mathcal{S}_{\mathcal{P}}(X) = (R_B(X), R_H(X), R_{\psi}(X)), \quad (57)$$

where R_B is the robust birth-window score over $\{B_5, B_6, B_7\}$, R_H is the robust high-support score over $\{S_8, S_9, O_{10+}\}$, and R_{ψ} is the robust strain score over $\{\psi_{\text{abs}}\}$.

The combined score

$$R_C(X) = R_{\mathcal{K}_C}(X)$$

answers whether the field is non-null relative to the selected same-density null model. The direction of $\mathcal{S}_{\mathcal{P}}(X)$ answers what kind of local Poole geometry carries the non-null signal.

Definition 5 (Normalized Poole geometry spectrum). *For non-null fields, define the normalized spectrum*

$$\Sigma_{\mathcal{P}}(X) = \frac{(R_B^2, R_H^2, R_{\psi}^2)}{R_B^2 + R_H^2 + R_{\psi}^2 + \varepsilon}. \quad (58)$$

The entries of $\Sigma_{\mathcal{P}}$ estimate the fractional birth-window, high-support, and strain contributions to the robust non-null signal.

Proposition 4 (Geometry-spectrometer principle). *Let X and X' be measured binary-field ensembles with the same active density. If their PooleQ/P channel-rate vectors differ after same-density null calibration, then they may separate either by magnitude, through R_C , or by direction, through $\Sigma_{\mathcal{P}}$. Thus PooleQ/P can distinguish fields that have the same low-order active density but different local organization, such as chain-like, branch-like, burst-like, or sheet-like support.*

Interpretive proof. The score R_C is computed from the non-redundant channel set $\mathcal{K}_C = \mathcal{K}_B \cup \mathcal{K}_H \cup \mathcal{K}_{\psi}$. If two fields have the same active density, density alone cannot explain a stable separation

from an exact same-density shuffled control. A high R_C therefore indicates non-null local channel structure relative to the null. The decomposition into R_B , R_H , and R_ψ records which disjoint channel groups account for that non-null structure. Normalizing their squared robust components gives Eq. (58), which removes overall magnitude and keeps the direction of the signal in Poole geometry space. $\square$

In the v5.5 run, this interpretation gives the following diagnostic map. IID and shuffled controls are null-like because $R_C < 2$. Straight lines and branching chains are mixed-geometry cases. Random-walk chains, compact bursts, and sheets are high-support-led cases. These labels are not universal physical categories; they are benchmark-calibrated names for regions of the synthetic Poole geometry space.

Class	R_C	Birth share	High-support share	Strain share	Spectrometer readout
IID null	0.28	0.64	0.01	0.36	null-like
Straight lines	5.28	0.44	0.17	0.39	mixed geometry
Random-walk chains	7.88	0.21	0.51	0.28	high-support-led
Branching chains	6.61	0.34	0.31	0.35	mixed geometry
Compact bursts	9.19	0.11	0.59	0.30	high-support-led
Sheets	9.76	0.13	0.58	0.30	high-support-led

Table 5: v5.6 spectrometer interpretation of the v5.5 decomposed benchmark. Shares are interpreted only after the field is non-null by R_C ; the IID row is shown for completeness but remains null-like.

The practical consequence is that PooleQ/P can be tested as a first-pass diagnostic layer. A future v5.6 verifier can use $\mathcal{S}_P(X)$ or $\Sigma_P(X)$ for a simple nearest-centroid geometry classifier. Such a classifier would not establish quantum decoding superiority by itself, but it would test whether PooleQ/P signatures can consistently separate null-like, chain-like, branch-like, burst-like, and sheet-like measured-field geometries under matched controls.

16 Interpretation and updated research program

The v5.6 interpretation supports the following updated claim:

PooleQ/P maps measured binary fields into an interpretable local-geometry signature space. Under exact active-density matching, structured fields remain separated from IID and same-density shuffled controls, while the vector (R_B, R_H, R_ψ) identifies whether the non-null signal is birth-window-led, high-support-led, strain-led, or mixed.

This result does *not* prove that PooleQ/P captures unknown quantum states. It does *not* prove quantum decoder superiority. It does *not* prove physical quantum advantage. It does justify a focused measurement-side claim: PooleQ/P is an interpretable local-geometry feature transform for p-bit fields, readout histories, and syndrome-event data.

The updated PooleQ/P program should now proceed in seven concrete steps.

1. **Geometry-signature classifier.** Build a simple nearest-centroid classifier in (R_B, R_H, R_ψ) or Σ_P space to test whether unknown synthetic measured fields can be labeled as null-like, chain-like, branch-like, burst-like, or sheet-like under matched controls.
2. **Paper-locked v5.5/v5.6 verifier.** Keep the v5.5 Colab verifier as the canonical reproduction script for this amended paper.

3. **Syndrome-like benchmark hardening.** Add temporal error chains, open boundary conditions, erasures, correlated bursts, and measurement-noise flips.
4. **Matched-null controls.** Keep exact active-count shuffling as a required control. Add site-wise marginal nulls when data are spatially inhomogeneous.
5. **Decoder augmentation test.** Compare a baseline classifier or decoder against the same model augmented with PooleQ/P decomposed features.
6. **PooleQ/p sampler.** Implement Eq. (17) and compare IID, Ising-like, and Poole-biased p-bit fields.
7. **Footprint-gate engineering verifier.** Continue checking raw AND, OR, NAND, NOR, and channel-resolved full-adder footprints, including routing, reset, fanout, and interference radius.

17 Conclusion

PooleQ/P Capture is a channel-resolved transform for binary stochastic and measured quantum-derived data. Its core mathematical components are

$$X \mapsto \Phi(X) = (B_5, B_6, B_7, S_5, \dots, S_9, O_{10+}, \psi), \quad (59)$$

$$q_i = (1 - p_i)C_i^{5:7} + p_iC_i^{5:9}, \quad (60)$$

$$B_7 = \text{Exact}_1, \quad B_6 = \text{Exact}_2, \quad B_5 = \text{Exact}_3 \quad (61)$$

for biased defect footprints, plus the decomposed null-calibrated metrics R_B , R_H , R_ψ , and R_C , and the geometry signature vector $\mathcal{S}_P = (R_B, R_H, R_\psi)$.

The amended v5.6 contribution is not a new overclaim. It is an interpretability hardening of the PooleQ/P capture idea into a geometry-spectrometer framing. The v5.5 experiment shows that PooleQ/P detects structured local correlation beyond active density alone, because structured classes remain high while exact same-density shuffled controls collapse to the IID baseline. The v5.6 interpretation adds that the decomposed vector $\mathcal{S}_P$ identifies the direction of the signal in local geometry space: birth-window geometry, high-support geometry, strain geometry, or a mixed combination. This makes PooleQ/P more suitable to present as an interpretable measurement-side feature layer and diagnostic preprocessor for p-bit fields and quantum syndrome histories.

A Machine-checkable footprint logic

The biased raw-gate and full-adder results can be checked by the following minimal Python logic:

```
from itertools import product

def raw_gate(bits, c):
    r = c + sum(bits)
    return int(2 <= r <= 4)

# Two-input raw gates
for A, B in product([0, 1], repeat=2):
    assert raw_gate([A, B], 0) == (A & B)
    assert raw_gate([A, B], 1) == (A | B)
    assert raw_gate([A, B], 3) == (1 - (A & B))
```



```

assert raw_gate([A, B], 4) == (1 - (A | B))

# Channel-resolved full-adder with one bias defect
for A, B, C in product([0, 1], repeat=3):
    s = A + B + C
    r = s + 1
    B7 = int(r == 2)
    B6 = int(r == 3)
    B5 = int(r == 4)
    SUM = B7 or B5
    CARRY = B6 or B5
    assert SUM == ((A + B + C) % 2)
    assert CARRY == int(A + B + C >= 2)

```

B Glossary

B_k Birth channel: inactive center with exactly k active neighbors.

S_k Survival channel: active center with exactly k active neighbors.

ψ Signed support strain; negative for under-support and positive for over-support.

O_{10+} Active cell with at least ten active neighbors; interpreted as overbulk/crowding.

PooleQ/P Channel-resolved Poole features applied to p-bit data or measured qubit-derived classical data.

Raw biased footprint gate The collapsed birth/no-birth output $Y_{n,c} = \mathbf{1}\{2 \leq c + \sum_j A_j \leq 4\}$.

Channel demultiplexing Any mechanism that converts B_5, B_6, B_7 labels into separate outputs rather than one collapsed live cell.

Null-calibrated capture A comparison of measured Poole channel rates against same-density IID or shuffled controls.

R_B Robust birth-window score using $\{B_5, B_6, B_7\}$.

R_H Robust high-support score using $\{S_8, S_9, O_{10+}\}$.

R_ψ Robust strain score using ψ_{abs} .

R_C Combined non-redundant robust score using birth, high-support, and strain groups; P_{active} is excluded to avoid channel redundancy.

$\mathcal{S}_P$ Poole geometry signature vector (R_B, R_H, R_ψ) .

Σ_P Normalized Poole geometry spectrum, using squared component shares of R_B , R_H , and R_ψ .

Geometry spectrometer The v5.6 interpretation of PooleQ/P as a map from measured binary fields into an interpretable local-geometry signature space.

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